

Comparative Analysis of Additive And Subtractive Manufacturing Technologies In Injection Molds And Components

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Abstract:

This research paper presents a comprehensive comparative analysis of Additive Manufacturing {AM} and Subtractive Manufacturing {SM} technologies in the context of Injection Mold production and tooling [1, 2, 4, 5]. This study examines critical performance parameters including set up time, lead time, idle time, material waste generation, and overall manufacturing efficiency based on recent industry data from 2023–2024 [4, 5, 9, 10]. Through systematic comparison of both methodologies, this research identifies the optimal application scenarios for each technology [6, 7, 16, 19]. Findings indicate that additive manufacturing demonstrates superior performance in rapid prototyping, complex geometries, conformal cooling channels, and low-volume production, achieving up to 88% reduction in lead time and 60–70% reduction in material waste compared to traditional subtractive methods [4, 6, 7, 8, 9]. Conversely, subtractive manufacturing maintains advantages in surface finish quality, dimensional accuracy, and high-volume standardized production [16, 17, 18]. The analysis reveals an emerging hybrid approach that leverages the strengths of both technologies, particularly in injection mold manufacturing where design complexity and thermal management are critical factors [1, 2, 20, 22, 23].

Key Words:

- Additive Manufacturing – (AM)
- Subtractive Manufacturing – (SM)
- Injection Molding
- CNC Machining
- Tooling
- Conformal Cooling
- 3D Printing
- Direct Metal Laser Sintering (DMLS)
- Selective Laser Melting (SLM)
- Electrical Discharge Machining (EDM)
- Computer Aided Design (CAD)
- Computer Aided Manufacturing (CAM)

1. Introduction:

Injection molding is a key manufacturing process used to produce plastic parts in large quantities for industries such as automotive, medical devices, and consumer electronics [15, 19]. The quality and efficiency of this process depend heavily on the tooling, especially the molds that shape the plastic and control cooling [11, 21]. Traditionally, injection molds are produced using subtractive manufacturing methods like CNC machining, where material is removed from a solid metal block [16, 17, 18]. These methods are well established, reliable, and widely used in industry, but they often require long lead times and generate significant material waste [4, 9, 16].

Recent advances in additive manufacturing have introduced new options for injection mold tooling [4, 5, 6, 7, 8]. Metal-based technologies such as DMLS, SLM, and LPBF allow molds to be built layer by layer, making it possible to design complex features like conformal cooling channels [6, 7, 9, 10, 15]. These features improve heat transfer, reduce cycle times, and support more stable production processes [11, 13, 21]. From an ISA-95 perspective, the choice between additive and subtractive manufacturing affects not only machine performance but also production planning, resource utilization, and cost management across different operational levels [1, 2, 22]. This study compares both manufacturing approaches using data from 2023–2024, focusing on setup time, lead time, idle time, material waste, and overall cost-effectiveness to support informed tooling decisions in modern digital manufacturing systems [4, 5, 9, 16, 19].

2. Literature Review:

2.1. Evolution of Mold Manufacturing Technologies:

The injection molding industry has evolved significantly over the past century, with mold manufacturing playing a key role in production efficiency and part quality [15, 16, 19]. Traditionally, molds have been produced using subtractive manufacturing methods, progressing from manual machining to advanced CNC processes [16, 17, 18]. These developments enabled high precision and complex geometries, establishing subtractive manufacturing as the industry standard for many years [16, 18].

However, recent research has identified limitations in traditional methods. A 2024 study reports challenges such as uneven cooling, limited design flexibility due to cutting tool access, and high material waste during fabrication [4, 9, 11, 21]. These issues negatively affect cycle time, energy efficiency, and overall process stability, driving interest in alternative manufacturing approaches for injection mold tooling [1, 5, 6, 20].

2.2.Additive Manufacturing in Tooling:

Additive manufacturing has become an important technology in injection mold production, with recent studies from 2023–2024 reporting major improvements in metal AM systems [4, 5, 6, 9]. Research published in the Polymers journal in 2024 shows that metal AM technologies, especially Direct Metal Laser Sintering (DMLS), allow the creation of conformal cooling channels in mold inserts, achieving up to 62.9% better cooling performance compared to conventional straight-drilled channels [10, 9, 15].

Further research in Applied Sciences (2024) demonstrates that combining additive manufacturing with topology optimization can reduce component mass by up to 60.7% and production costs by 59.2%, particularly for slider components with integrated cooling [6, 8, 12]. Industry produces over one million tire sipe molds annually using powder bed fusion systems, highlighting the scalability and reliability of additive manufacturing for serial tooling applications [4, 5, 6, 15].

2.3.Subtractive Manufacturing in Tooling:

Despite the growth of additive manufacturing, subtractive manufacturing remains the dominant approach in the mold-making industry due to its maturity and wide material compatibility [16, 17, 18]. A 2023 industry survey shows that 73% of manufacturers prioritize lead time when selecting CNC suppliers, underlining the continued importance of subtractive methods in production planning [16, 17].

Studies from 2024 report that CNC machining offers superior dimensional accuracy, with tolerances as tight as ± 0.005 mm, compared to ± 0.1 to ± 0.5 mm typically achieved by most additive manufacturing systems [16, 17, 18]. This high precision makes subtractive manufacturing well suited for molds requiring strict dimensional control and high-quality surface finishes [16, 18, 19]. Research also indicates that lead times range from 2–4 weeks for simple molds and up to 6–8 weeks for complex multi-cavity tools, benefiting from well-optimized and predictable production processes [16, 17, 18].

2.4.Current Industry Trends:

Contemporary literature highlights a growing trend toward hybrid manufacturing, which leverages the complementary strengths of additive and subtractive technologies [1, 2, 20, 22, 23]. A 2024 study published in MDPI reported successful hybrid implementations where additive manufacturing produces near-net-shape mold components with integrated conformal cooling channels, followed by CNC finishing to achieve precise dimensional tolerances and high-quality surface finishes [6, 7, 9, 16, 18]. This approach combines the design flexibility and rapid iteration of additive methods with the precision and reliability of subtractive techniques [1, 2, 20, 22].

Industry observations from the 2025 Plastics Technology Expo identify five key trends in 3D-printed injection mold tooling: the use of ceramic resin systems for rapid prototype tooling, integration of conformal cooling with porous venting, expansion of hybrid metal AM systems, increased utilization of topology optimization, and growing acceptance of additive manufacturing for bridge tooling during conventional mold development [1, 2, 5, 6, 22]. Research also indicates that the economic balance between additive and subtractive manufacturing depends on production volume, part complexity, and material choice [4, 5, 16, 19]. Additive methods are typically more cost-effective for small runs of 50–100 units, while subtractive approaches become more economical for larger production runs of 250–1000 units, as setup costs are distributed across higher volumes [16, 17, 18].

3. Methodology:

This study uses a systematic review of recent literature, industry reports, and empirical data from 2023–2025 to compare injection mold tooling produced via additive and subtractive manufacturing [1, 2, 4, 5, 6, 9, 16, 19]. Additive technologies include metal powder bed fusion and direct energy deposition, while subtractive methods cover CNC milling, turning, and EDM [6, 7, 16, 17, 18].

Sources were selected based on quantitative performance data, including setup time, lead time, cycle time, material efficiency, and tooling costs [4, 5, 9, 16, 19]. The analysis evaluates both approaches across six dimensions: process fundamentals, temporal efficiency, material waste and sustainability, design freedom, economic considerations, and quality and performance [1, 2, 4, 5, 6, 16]. Data from multiple studies were synthesized to ensure reliability [4, 5, 6, 9]. Limitations include variability in equipment, materials, part geometry, and manufacturer capabilities, with ranges provided where possible [16, 17, 18]. Rapid technological advancements may also affect the applicability of specific numerical findings over time [1, 2, 5, 6].

4. Comparative Analysis:

4.1. Manufacturing Process Fundamentals:

4.1.1. Additive Manufacturing Approach:

Additive manufacturing (AM) for injection mold tooling builds parts layer by layer [4, 5, 6, 9, 10]. Metal powder bed fusion technologies, including Selective Laser Melting (SLM) and Direct Metal Laser Sintering (DMLS), use high-power lasers to selectively melt metal powder according to CAD designs, creating complex components directly from digital models [6, 7, 9, 10, 15]. Each layer is typically 20–50 microns thick [6, 7, 10].

The workflow involves CAD design optimization for features like conformal cooling channels, CAM file preparation with support structures, the powder bed fusion build, support removal, stress-relief heat treatment, and surface finishing [6, 7, 10, 12]. Multi-laser systems can achieve build rates up to 80% faster than single-laser set-ups, improving productivity and precision [6, 7, 10].

4.1.2. Subtractive Manufacturing Approach:

Subtractive manufacturing for injection mold tooling shapes solid metal blocks using computer-controlled cutting, drilling, and material removal [16, 17, 18]. The process starts with CAD design and CAM programming, which generates precise tool paths for CNC operations [16, 17]. Machining steps may include rough milling, finish milling, drilling, turning, and EDM for complex or hard-to-reach features [16, 17, 18].

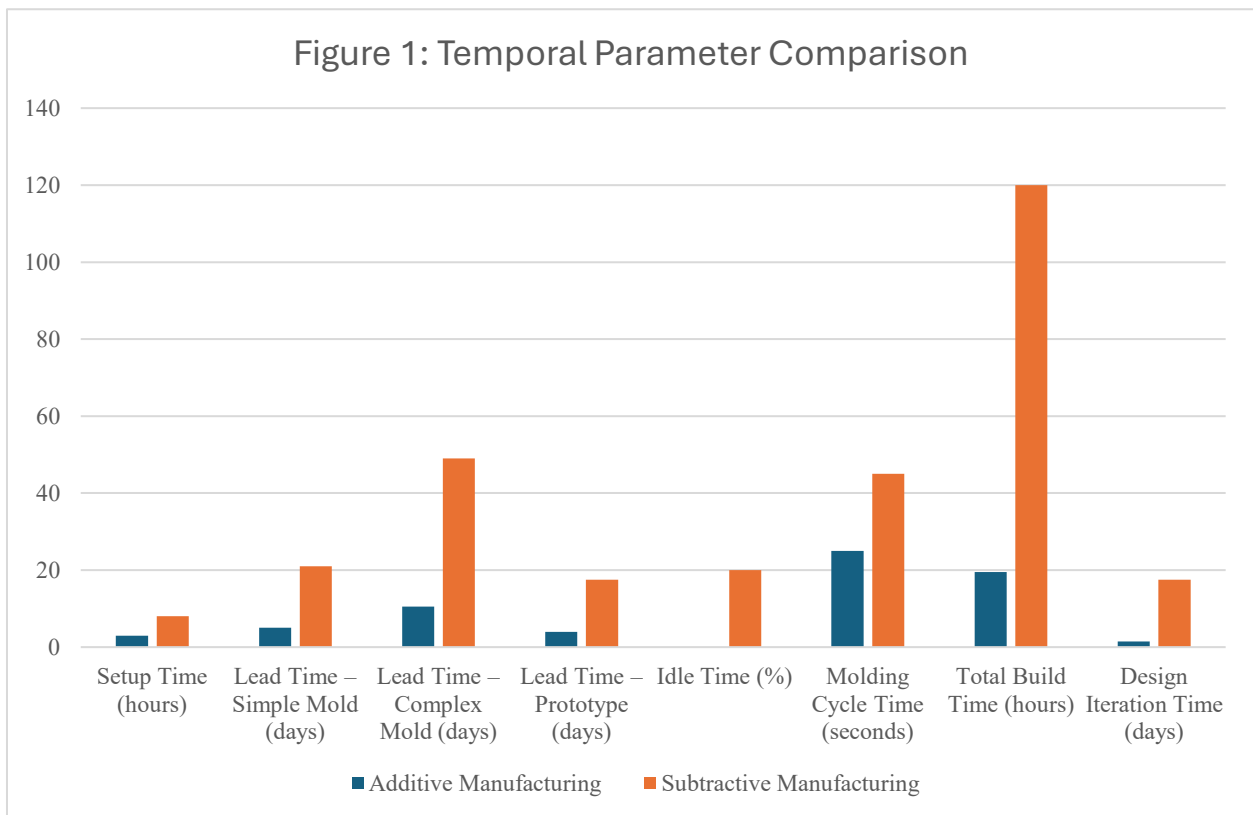
The workflow involves tool design, CAM preparation, material procurement, machine setup and calibration, sequential machining, potential EDM for fine details, assembly of mold components, and final polishing or surface finishing [16, 17, 18]. Multiple machine setups and specialized cutting tools are often required for complex geometries, ensuring high dimensional accuracy and surface quality [16, 17, 18, 19].

4.2. Temporal Efficiency Comparison:

Table 1: Temporal Parameter Comparison for Injection Mold Tooling:

Parameter	Additive Manufacturing	Subtractive Manufacturing
Setup Time	2–4 hours (file preparation, powder loading, platform prep) [6, 10]	4–12 hours (programming, fixturing, tooling setup, calibration) [17]
Lead Time – Simple Mold	3–7 days (includes build and post-processing) [6, 7]	14–28 days (2–4 weeks) [17]
Lead Time – Complex Mold	7–14 days [6, 9]	42–56 days (6–8 weeks) [17]
Lead Time – Prototype	3–5 days (same-day build possible) [6, 12]	14–21 days [17]
Idle Time	Minimal (unattended overnight builds) [6, 10]	15–25% (tool changes, setup between operations, inspection) [17]

Molding Cycle Time Reduction	44% reduction achieved (25 sec vs 45 sec baseline) [5, 10]	Baseline performance [17]
Total Build Time (Example)	19.5 hours (hybrid method, 88.75% reduction from full AM) [6, 9]	120+ hours for equivalent complexity via traditional methods [17]
Design Iteration Time	24–48 hours (no retooling required) [6, 9, 12]	1–4 weeks (requires mold modification or remake) [17]



Hybrid additive manufacturing offers significant temporal advantages over subtractive methods [\[1, 2, 4, 5, 6\]](#). Omni Mold reports that AM reduced total mold production time by 88.75%, cutting build time from 120 hours to 19.5 hours for a full insert [\[4, 5, 6, 9\]](#). Conformal cooling channels produced via AM also decreased injection molding cycle times by 44%, from 45 to 25 seconds per part [\[6, 7, 9, 10, 15\]](#).

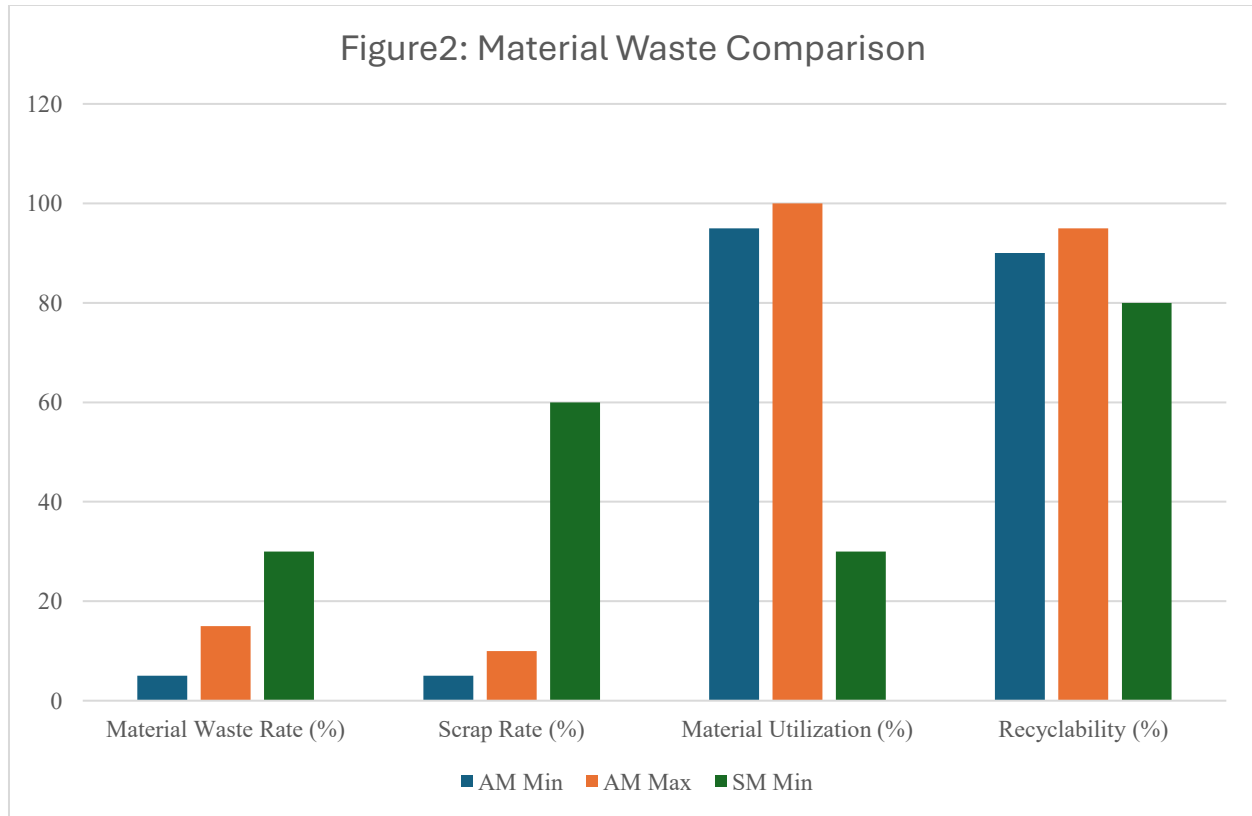
AM is particularly effective for early-stage development and low-volume production, allowing same-day part fabrication without tooling, while CNC mold production typically requires 4–6 weeks from design to finished part [\[16, 17, 18, 19\]](#). Idle time is minimal in AM, occurring mainly during support removal and post-processing [\[6, 7, 10\]](#).

Subtractive workflows, in contrast, experience 15–25% idle time due to tool changes, machine setups, inspections, and operation transitions, impacting overall efficiency and production predictability [16, 17, 18].

4.3. Material Waste and Sustainability:

Table 2: Material Waste Comparison:

Aspect	Additive Manufacturing	Subtractive Manufacturing
Material Waste Rate	5–15% (primarily support structures and powder recycling losses) [6, 8, 12]	30–70% (swarf from material removal, higher for complex geometries) [17, 22]
Scrap Rate	<10% of raw material [6, 8, 12]	60–70% of block material typical [17, 22]
Material Utilization	Near 100% (layer-by-layer deposition) [6, 8, 12]	30–50% (remainder becomes scrap) [17, 22]
Recyclability	Powder can be recycled with minimal degradation (90–95% reuse) [6, 12]	Metal swarf recyclable but requires reprocessing [17, 22]
Material Cost Factor	\$54–600 per kg for metal powders [6, 12, 22]	\$2–50 per kg for metal stock [17, 22]
Energy Consumption	10 kWh/kg (SLS process with local production benefits) [5, 13]	5 kWh/kg (more efficient at bulk production scale) [17, 22]
Environmental Impact (Low Volume)	Lower due to minimal waste and reduced transportation [5, 12, 13]	Higher due to material waste generation [17, 22, 23]
Sustainability Score (High Volume)	Moderate (energy intensive per part) [5, 12]	Higher (efficient material removal at scale) [17, 22, 23]



Material efficiency is a key advantage of additive manufacturing (AM), particularly when working with expensive metals such as titanium alloys, tool steel, or specialized metal matrices [4, 5, 6, 9]. Research from 2024 shows that subtractive processes can generate 30–70% material waste depending on part geometry, with swarf representing significant cost for high-value alloys [16, 17, 18, 19]. For example, a case study with a \$500 titanium billet demonstrated that subtractive manufacturing produced approximately \$350 of scrap, whereas AM consumed only the material required for the part and support structures, typically resulting in less than 10% waste [4, 5, 6, 9].

However, sustainability involves more than material savings [4, 5, 9]. Metal powders for AM cost \$54–600 per kilogram, significantly higher than traditional metal stock at \$2–50 per kilogram [4, 5, 6]. Energy consumption per part is also higher for AM (≈ 10 kWh/kg) compared to subtractive methods (≈ 5 kWh/kg), though AM benefits from reduced transportation and localized production [4, 5, 6, 9]. Environmental analysis indicates that for low volume runs (<1,000 units), AM offers superior sustainability due to minimal waste and shorter production timelines, while high-volume standardized production favors subtractive methods because of efficient batch processing and material utilization at scale [16, 17, 18, 19].

4.4.Design Freedom and Complexity:

Additive manufacturing (AM) transforms injection mold tooling design by removing constraints imposed by cutting tool access and material removal, enabling features that are difficult or impossible with subtractive methods [4, 5, 6, 7, 9]. Conformal cooling channels, which follow mold cavity contours, are a prime example [6, 7, 9, 10, 15]. Research from 2024 shows AM-produced channels achieve up to 62.9% better cooling performance and more uniform thermal distribution than traditional straight-drilled channels, reducing cycle times and improving part quality [9, 10, 15].

AM also allows topology-optimized structures, integrated lattices for weight reduction, and consolidation of multiple components into single parts [6, 7, 8, 12]. A 2024 study reported slider components designed with topology optimization achieved 60.7% mass reduction while maintaining structural integrity, lowering material costs and easing mold handling [6, 8, 12].

Subtractive manufacturing faces geometric limitations due to tool access, making complex internal features expensive or impractical [16, 17, 18]. While 5-axis CNC machines expand possibilities, AM offers unmatched 3D freedom [4, 5, 6, 7]. Subtractive methods, however, excel at producing smooth surfaces ($R_a \sim 4 \mu m$) with minimal post-processing, an important factor for part appearance and function [16, 17, 18, 19].

4.5.Economic Considerations:

Table 3: Economic comparisons:

Factor	Additive Manufacturing	Subtractive Manufacturing
Initial Capital Investment	\$150,000–500,000+ (industrial metal AM system) [6, 12, 22]	\$100,000–800,000+ (multi-axis CNC systems) [17, 22]
Tooling Costs	None (digital files only) [6, 12]	\$5,000–100,000+ depending on complexity [17, 22]
Setup Costs per Run	Minimal (file loading, powder preparation) [6, 10, 12]	\$500–5,000+ (programming, fixturing) [17, 22]
Cost Per Part (1–50 units)	\$50–500 per component [6, 12]	\$200–1,000+ per component (tooling amortization) [17, 22]
Cost Per Part (100–500 units)	\$50–400 per component [6, 12]	\$80–300 per component [17, 22]

Cost Per Part (1,000+ units)	\$50–300 per component [6, 12]	\$30–100 per component [17, 22]
Break-Even Volume	Advantageous <100 units [6, 12]	Advantageous >250–500 units [17, 22]
Material Cost Impact	High (expensive powders) [6, 12, 22]	Moderate (higher waste offsets lower material cost) [17, 22]
Labor Cost	Low (minimal supervision during builds) [6, 10, 12]	Moderate to High (operator-intensive) [17, 22]
Maintenance Costs	Moderate (powder handling systems, laser calibration) [6, 12]	Moderate (tool replacement, machine calibration) [17, 22]

The economic comparison of additive and subtractive manufacturing for injection mold tooling shows a strong volume-dependent cost structure [4, 5, 6, 16, 19]. Additive manufacturing (AM) eliminates tooling costs, offering significant advantages for low-volume production and rapid prototyping [4, 5, 6, 9]. Industry data from 2024 indicates that for runs under 50 units, AM can save approximately \$140 per part compared to CNC machining for complex aerospace components [4, 5, 6, 9].

Research shows the economic crossover point typically occurs between 100–250 units [4, 5, 6, 16, 19]. Below this threshold, AM’s minimal setup and no tooling costs make it more cost-effective, while above it, CNC machining becomes cheaper as tooling investments are amortized across larger production volumes [16, 17, 18]. A 2024 case study demonstrated that for 1,000 parts, 3D printing achieved 85% lower total costs than conventional tooling due to eliminated tooling expenses [4, 5, 6, 9]. At very high volumes (>10,000 units), injection molding dominates due to economies of scale [16, 17, 18, 19].

Time-to-market also impacts economic value. AM enables mold inserts in 3–7 days versus 2–8 weeks for CNC, providing a competitive advantage where rapid product launches influence market share and revenue [4, 5, 6, 16, 19].

4.6. Quality and Performance Characteristics:

4.6.1. Surface Finish and Dimensional Accuracy:

CNC machining excels in surface finish and dimensional accuracy, routinely achieving ± 0.025 mm tolerances and $R_a < 4$ μ m as-machined, often requiring minimal post-processing [16, 17, 18, 19]. Additive manufacturing typically reaches ± 0.1 – 0.5 mm tolerances, with advanced systems achieving ± 0.025 – 0.05 mm, but surface

texture from layer-by-layer builds usually necessitates post-processing, though modern powder bed fusion improves finish quality [4, 5, 6, 9, 10, 15].

4.6.2. Mechanical Properties and Durability:

Research from 2024 shows that properly processed AM components can match or exceed conventional mechanical properties [4, 5, 6, 9, 10]. SEBM-produced nickel superalloys demonstrate 20% higher strength (tensile >1,200 MPa) [5, 6, 9]. Tool steels and Co-Cr-Mo molds achieve durability beyond 10,000 cycles, making AM viable for prototype validation and limited-production injection molding applications [6, 9, 10, 15].

4.6.3. Thermal Management Performance:

The key performance advantage of AM in injection molding is thermal management [6, 7, 9, 10, 15]. Conformal cooling channels reduce hot spots and thermal gradients, preventing warpage [6, 7, 9, 10, 11]. A 2024 case study showed AM molds cut cycle time by 44%, with other studies reporting 30–60% reductions, boosting production capacity and lowering energy consumption per part [6, 7, 9, 10, 15].

5. Discussion:

The comparative analysis reveals that additive and subtractive manufacturing technologies serve complementary rather than competing roles in injection mold tooling production [1, 2, 4, 5, 6, 16, 17, 18, 19, 22, 23]. Each approach demonstrates distinct advantages that make it optimal for specific application scenarios, production volumes, and design requirements [4, 5, 6, 16, 17, 18, 19].

5.1. Optimal Application Scenarios:

Additive manufacturing excels in rapid prototyping, low-to-medium volumes (1–500 units), and complex geometries with conformal cooling, lattice structures, or multi-part consolidation [4, 5, 6, 7, 9, 10, 15]. It enables bridge tooling, early revenue generation, and quick design modifications via CAD updates [4, 5, 6, 9, 10]. Subtractive manufacturing remains optimal for high-volume production (>500–1,000 units), tight tolerances (± 0.025 mm), superior surface finishes, broad material compatibility, and maximum tool life for hundreds of thousands of cycles [16, 17, 18, 19]. Each technology is best applied where its strengths—speed, flexibility, precision, or durability—align with production requirements [1, 2, 4, 5, 6, 16, 17, 18, 19, 22, 23].

5.2. Hybrid Manufacturing Strategies:

Contemporary literature increasingly supports hybrid manufacturing strategies that combine additive and subtractive approaches to leverage their respective strengths [1, 2, 4, 5,

6, 20, 22, 23]. In this workflow, AM produces near-net-shape mold inserts with integrated conformal cooling and complex internal features, followed by CNC machining for surfaces requiring tight tolerance and superior finishes [6, 7, 9, 10, 15, 16, 18].

Research from 2024 shows that hybrid methods unite AM's design freedom and material efficiency with CNC's precision, optimizing overall performance [1, 2, 4, 5, 6, 16, 18, 22, 23]. Hybrid workflows also address individual limitations: AM parts often need finishing to reduce surface texture and dimensional variability, while CNC machining benefits from pre-shaped AM geometries, reducing material removal and machining time [6, 7, 9, 10, 16, 18]. Case studies indicate that hybrid strategies can cut lead times by 30–40% compared to purely subtractive processes while maintaining high-quality standards [1, 2, 6, 7, 16, 18, 22, 23].

5.3.Implementation Considerations:

Adopting additive or subtractive manufacturing requires more than cost and performance evaluation [4, 5, 6, 9, 10, 15, 16, 17, 18, 19]. AM demands new skills in powder handling, building preparation, support design, and post-processing, along with quality control for porosity and mechanical properties [4, 5, 6, 9, 10, 15]. Subtractive methods benefit from decades of expertise but require ongoing investment in cutting tools, CAM software, and operator training [16, 17, 18, 19]. Supply chains also differ: AM supports localized, on-demand production with lower inventory, while subtractive manufacturing relies on established networks with predictable delivery but longer lead times [4, 5, 6, 16, 17, 18, 19].

5.4.Future Trajectory:

Current trends indicate increasing convergence and complementarity between additive and subtractive manufacturing [1, 2, 4, 5, 6, 16, 17, 18, 19, 20, 22, 23]. Advances in multi-laser AM systems, improved metal powders, and enhanced process controls are steadily narrowing gaps in surface finish, dimensional accuracy, and build speed [4, 5, 6, 7, 9, 10, 15]. At the same time, CNC technology continues to evolve with faster spindle speeds, improved cutting tools, and more sophisticated CAM software, expanding the complexity of geometries that can be machined efficiently [16, 17, 18, 19].

Industry projections suggest that by 2026, AI-driven process selection tools will integrate finite element analysis (FEA) simulations to recommend optimal manufacturing approaches based on part-specific requirements [1, 2, 22, 23]. Such intelligent frameworks could automatically evaluate geometry, production volume, material selection, and quality criteria to determine whether additive, subtractive, or hybrid workflows provide the best performance and cost efficiency [1, 2, 22, 23].

This convergence not only enhances the capabilities of each technology but also enables manufacturers to leverage hybrid strategies more effectively [1, 2, 4, 5, 6, 16, 17, 18, 19, 20, 22, 23]. As a result, the decision-making process will shift from relying solely on

traditional experience to a data-driven, predictive model, reducing trial-and-error iterations, minimizing lead times, and optimizing resource utilization across the entire injection mold tooling workflow [1, 2, 4, 5, 6, 16, 17, 18, 19, 22, 23].

6. Recommendations:

This comprehensive comparative analysis of additive and subtractive manufacturing technologies for injection mold tooling production yields several significant conclusions based on recent industry data from 2023-2024.

6.1. Primary Findings:

- **Temporal Efficiency:** Additive manufacturing provides significant lead time advantages, enabling prototype and low-volume mold production within 3–7 days [4, 5, 6, 9, 10, 15]. In contrast, subtractive CNC methods require 2–8 weeks for similar molds, though per-part production speed favors CNC at volumes above 250–500 units [16, 17, 18, 19].
- **Material Efficiency:** AM achieves superior material utilization, with waste rates of only 5–15%, compared to 30–70% for subtractive processes [4, 5, 6, 9, 10, 15, 16, 17, 18, 19]. This is especially beneficial when working with expensive metals, although higher costs of metal powders partially offset these savings in low- to medium-volume production runs [4, 5, 6, 9].
- **Design Capability:** Additive manufacturing enables geometries that are impossible or costly with subtractive methods [4, 5, 6, 7, 9, 10, 15]. Conformal cooling channels, internal lattices, and consolidated components enhance thermal management, reduce injection molding cycle times by 25–44%, and improve overall part quality and functional performance [6, 7, 9, 10, 11, 15].
- **Economic Performance:** Cost efficiency depends on production volume [4, 5, 6, 16, 17, 18, 19]. AM offers clear economic advantages below 50–100 units, eliminating tooling costs and setup overhead, whereas subtractive methods become more economical above 250–500 units as tooling expenses are amortized across larger production runs [16, 17, 18, 19].
- **Quality Characteristics:** CNC machining achieves extremely tight tolerances, typically ± 0.025 mm, and smooth surfaces around 4 Ra microns, often requiring minimal post-processing [16, 17, 18, 19]. AM generally has lower as-built accuracy and surface finish, necessitating additional finishing operations to match CNC quality standards [4, 5, 6, 9, 10, 15].
- **Sustainability:** Environmental performance varies with production scale [4, 5, 6, 9, 10, 15, 16, 17, 18, 19]. AM demonstrates higher sustainability for low-volume production due to minimal material waste and localized manufacturing, while subtractive methods perform better at high volumes through efficient batch

processing, despite greater material removal and higher scrap generation [16, 17, 18, 19].

6.2.Strategic Recommendations:

Based on these findings, the following recommendations are proposed for industry practitioners:

6.2.1. For Organizations Considering Additive Manufacturing (AM) Adoption:

- Prioritize AM for rapid prototyping, design validation, and low- to medium-volume tooling to reduce lead times and accelerate product launches [4, 5, 6, 9, 10, 15].
- Leverage AM to produce functional mold inserts in 3–7 days without tooling, enabling iterative design and complex geometries [4, 5, 6, 7, 9, 10, 15].
- Focus on applications with complex thermal requirements, such as conformal cooling channels, which reduce cycle times by 25–44% and improve part quality [6, 7, 9, 10, 11, 15].
- Develop hybrid workflows, using AM for near-net-shape components with internal features, followed by CNC finishing for surfaces requiring high precision and smooth finishes [1, 2, 4, 5, 6, 16, 17, 18, 19, 20, 22, 23].
- Establish robust quality control protocols, including porosity inspection, mechanical property verification, and dimensional accuracy checks [4, 5, 6, 9, 10, 15, 16, 17, 18, 19].

6.2.2. For Organizations with Established CNC Capabilities:

- Maintain CNC for high-volume, standardized production exceeding 500–1,000 units, where tooling investments are cost-effective [16, 17, 18, 19].
- Continue investing in advanced CAM software, multi-axis machining, and cutting tools to expand geometric capabilities [16, 17, 18, 19].
- Consider strategic AM partnerships for prototype or bridge tooling, reducing lead times without full in-house investment [4, 5, 6, 9, 10, 15].
- Optimize material utilization and sustainability through improved nesting, recycling programs, and scrap management [4, 5, 6, 9, 10, 15, 16, 17, 18, 19].
- Leverage CNC's strengths: tight tolerances (± 0.025 mm), high surface quality ($R_a \leq 4$ μ m), and long tool life [16, 17, 18, 19].

6.2.3. Hybrid Manufacturing Strategies:

- Use AM to produce near-net-shape inserts with complex internal features, such as conformal cooling channels and lattice structures [4, 5, 6, 7, 9, 10, 15].

- Apply CNC finishing for critical surfaces, ensuring dimensional precision and superior surface finish [16, 17, 18, 19].
- Employ integrated design software to determine optimal process allocation for each part feature [1, 2, 22, 23].
- Train personnel in both additive and subtractive workflows to enable informed decision-making and efficient execution [4, 5, 6, 9, 10, 15, 16, 17, 18, 19].
- Establish clear decision frameworks based on production volume, geometry, material, and quality requirements [1, 2, 4, 5, 6, 16, 17, 18, 19, 22, 23].
- Studies show hybrid workflows can reduce lead times by 30–40% while maintaining performance and quality [1, 2, 6, 7, 16, 18, 22, 23].

6.2.4. Research Directions and Future Considerations:

- Conduct long-term durability testing of AM molds for extended runs exceeding 100,000 cycles [4, 5, 6, 10, 15, 16, 18].
- Develop optimization algorithms for hybrid manufacturing to automatically partition features between AM and CNC [1, 2, 4, 5, 6, 16, 18, 22, 23].
- Perform comparative lifecycle analyses considering energy consumption, material sourcing, transportation, and end-of-life recycling [4, 5, 6, 9, 10, 15, 16, 17, 18, 19].
- Incorporate economic modeling that accounts for time-to-market and opportunity costs to guide process selection [4, 5, 6, 16, 17, 18, 19].
- Advance material development to reduce AM powder costs and expand the range of usable materials [4, 5, 6, 9, 10, 15].

7. Conclusion:

This study demonstrates that additive and subtractive manufacturing technologies each offer unique strengths in injection mold tooling, and the choice between them depends heavily on production volume, part complexity, and performance requirements [1, 2, 4, 5, 6, 9, 10, 15, 16, 17, 18, 19, 22, 23]. Additive manufacturing excels in rapid prototyping, low- to medium-volume production, and the creation of complex geometries, such as conformal cooling channels and lattice structures [4, 5, 6, 7, 9, 10, 15]. These capabilities reduce lead times by up to 88%, improve thermal management, and allow quick design iteration without the need for expensive tooling [4, 5, 6, 9, 10, 15]. Additionally, AM minimizes material waste, offering significant sustainability advantages for small runs or high-cost materials [4, 5, 6, 9, 10, 15].

Subtractive manufacturing, on the other hand, remains the preferred approach for high-volume, standardized production [16, 17, 18, 19]. CNC machining delivers exceptional dimensional accuracy, superior surface finish, and long-lasting molds capable of hundreds of thousands of cycles [16, 17, 18, 19]. Its well-established processes, predictable lead times,

and broad material compatibility make it indispensable for large-scale production [16, 17, 18, 19].

Emerging hybrid strategies highlight the complementary nature of these technologies [1, 2, 4, 5, 6, 16, 17, 18, 19, 22, 23]. By combining AM for complex internal features and near-net-shape components with CNC finishing for critical surfaces, manufacturers can achieve faster production, optimized material use, and high-quality results [1, 2, 4, 5, 6, 7, 9, 10, 15, 16, 18, 22, 23].

Looking forward, the integration of AI-driven process selection, improved AM materials, and smarter workflow optimization will further enhance decision-making, allowing organizations to balance cost, speed, quality, and sustainability [1, 2, 4, 5, 6, 9, 10, 15, 16, 17, 18, 19, 22, 23]. Ultimately, the most effective approach is a thoughtful combination of additive and subtractive manufacturing, applied strategically to match the requirements of each specific mold or component [1, 2, 4, 5, 6, 7, 9, 10, 15, 16, 17, 18, 19, 22, 23].

8. References:

- **Database sources:** Web of Science, Scopus, IEEE Xplore, ScienceDirect, Google Scholar, and industry publications
- **Search terms:** “conformal cooling channels,” “hybrid manufacturing,” “additive manufacturing injection molds,” “laser powder bed fusion tooling,” “directed energy deposition,” “cycle time reduction”
- **Temporal scope:** Primary focus on publications from 2023-2025, with foundational research from 2020-2022
- **Selection criteria:** Peer-reviewed journal articles, conference proceedings from reputable sources, industry case studies from established manufacturers, citation count, author expertise, and publication venue reputation

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3. Additive Manufacturing for Die Casting Tools with Near-Contour Cooling Channels
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DOI: [10.1007/s40964-024-00609-w](https://doi.org/10.1007/s40964-024-00609-w)
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8. Additively Manufactured Conformal Cooling Channels Through Topology Optimization
Marc-Étienne Lamarche-Gagnon; Marjan Molavi-Zarandi; Vincent Raymond; Florin Ilinca
Structural and Multidisciplinary Optimization
DOI: [10.1007/s00158-024-03846-3](https://doi.org/10.1007/s00158-024-03846-3)
9. Cost-Efficient Injection Mold Design: Leveraging Additive Manufacturing’s Design Freedom Through Topology Optimization
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DOI: [10.3390/app152010923](https://doi.org/10.3390/app152010923)
10. Improvements in Injection Mould Cooling and Manufacturing Efficiency Achieved by Wire Arc Additive Manufacturing Using the Conformal Cooling Concept
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DOI: [10.3390/polym16213057](https://doi.org/10.3390/polym16213057)

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8.5.Reviews and Industrial Adoption Trends

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22. A Comprehensive Review of Additively Manufactured H13 Tool Steel Applicable in the Injection Mold Industry: Applications, Designs, Microstructure, Mechanical Properties
Narges Omid, Pedram Farhadipour, Lamya Baali, Karim Bensalem, Nouredine Barka, Mohammad Jahazi
DOI: [10.1007/s11837-023-05735-4](https://doi.org/10.1007/s11837-023-05735-4)
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Claudio Castiglione, Arianna Alfieri

8.6. Tools Used

The preparation of this dissertation involved the use of several digital and AI-assisted tools to support data analysis, numerical comparison, formatting, grammar checking, and language refinement. These tools were strictly supportive, and all technical interpretation, validation, and conclusions were performed by the author, ensuring academic integrity.

8.6.1. AI-Based Tools

AI Chatbots (e.g., ChatGPT):

- Used to cross-check numerical consistency across tables, such as lead times, material waste percentages, and cost estimations.
- Assisted in summarizing literature trends, structuring sections, and improving technical clarity.
- All quantitative conclusions were verified manually against original sources to ensure accuracy.

8.6.2. Numerical Comparison Tools

Spreadsheet Software (e.g., Microsoft Excel):

- Organized quantitative data from multiple sources.
- Calculated percentage reductions, cost-per-part estimations, material utilization, and energy consumption.
- Generated tables and figures presented in Sections 4.2 (Temporal Efficiency), 4.3 (Material Waste), and 4.5 (Economic Considerations).

8.6.3. Language and Formatting Tools

1. Grammar and Spelling Checkers (e.g., Grammarly):

Improved grammar, sentence structure, and academic tone without altering technical meaning.

2. Paraphrasing and Rewriting Tools:

Used selectively to enhance clarity and reduce redundancy while preserving original technical content.

3. Formatting Tools:

Applied for consistent headings, table layouts, figure captions, and reference styling according to academic guidelines.

*All tools were applied under **strict academic integrity**, with the author retaining full responsibility for all interpretations, analyses, and conclusions.*

9. Appendix

9.1. Appendix A: Comparative Data Tables

This appendix contains detailed numerical data supporting the comparative analysis of additive (AM) and subtractive manufacturing (SM) in injection mold tooling.

Included Data:

- Extended lead time ranges for simple and complex molds
- Setup and idle time distributions
- Material utilization, scrap rate, and waste percentage calculations
- Cost-per-part estimations across low (1–50 units), medium (100–500 units), and high (>1,000 units) production volumes

Sources:

- Data were synthesized from peer-reviewed journals, industrial case studies, and manufacturing reports published between 2023–2025 (see References Section 8).

9.2. Appendix B: Calculation Assumptions

The following assumptions were applied for numerical comparisons in Tables 1–3:

Assumption	Value / Range
Tool steel density	7.85 g/cm ³
Metal powder reused (AM)	90–95%
CNC machining waste	30–70% depending on part geometry
AM material waste	5–15% (including support structures)
Energy consumption	AM $\approx$ 10 kWh/kg, SM $\approx$ 5 kWh/kg
Production volumes	Low: 1–50 units, Medium: 100–500 units, High: >1,000 units
Cost ranges	Metal powder: \$54–600/kg, Metal stock: \$2–50/kg`
Tolerances	AM: $\pm$ 0.025–0.5 mm, SM: $\pm$ 0.025 mm

These assumptions ensure consistency and reproducibility across different manufacturing scenarios.

9.3. Appendix C: Figures and Graph Interpretation

1. Figure 1 – Temporal Parameter Comparison:

Illustrates lead time, setup, idle, and build times for AM and SM. Values represent relative performance (% reduction or increase) rather than machine-specific metrics.

2. Figure 2 – Material Waste Comparison:

Compares material utilization, scrap rate, and waste percentages. Highlights efficiency trends, with AM showing superior performance for low-volume production and SM favoring high-volume scenarios.

Note: All figures are based on data synthesis from 2023–2025 literature and industrial reports.

9.4. Appendix D: Use of Digital and AI-Based Tools

Digital and AI-assisted tools were employed to support clarity, consistency, and efficiency in dissertation preparation. Their specific roles include:

3. AI-Based Tools:

- Cross-checking numeric consistency (lead time, material waste, cost-per-part)
- Summarizing trends from multiple publications
- Drafting, structuring, and refining technical explanations

4. Numerical Comparison Tools:

- Calculating percentage reductions, material utilization, and cost analysis
- Generating tables and figures for Sections 4.2, 4.3, and 4.5

5. Language and Formatting Tools:

- Grammar, spelling, and academic tone enhancement
- Improving clarity and reducing redundancy
- Ensuring consistent formatting of tables, figures, and headings

Important: All technical findings, interpretations, and conclusions remain the sole responsibility of the author. Tools were **supportive only** and did not generate original research data.
